

AI-Assisted Threat Analysis Log

Segment 3 — traffic_segment_3.csv

1. AI Tool Used

Tool: ChatGPT

ChatGPT was used as an AI-assisted security analysis tool to review the PCAP-derived CSV dataset and surface potentially suspicious patterns — communication relationships, protocol usage, and unusual traffic behaviour worth a closer look.

2. Dataset Information

Dataset analyzed: traffic_segment_3.csv

Source: Extracted from 2021-08-16-formbook.pcap

Records analyzed: 1,188 packets

The analysis set out to identify:

- Frequently communicating IP addresses
- Internal and external communication relationships
- Repeated communication patterns
- Potential automated behaviour
- Port and protocol usage
- Data transfer patterns
- Potential security concerns

3. AI Prompt Used

The following prompt was submitted to ChatGPT to begin the traffic investigation:

You are assisting with network traffic analysis as a cybersecurity analyst.

I have provided a CSV file extracted from a PCAP network capture.
This is Segment 3 of a larger network traffic investigation.
Analyze this dataset and identify potential security concerns.

Perform traffic investigation focusing on:

1. Identify the most frequently communicating source and destination IP addresses.
2. Determine internal/private and external/public IP addresses using standard IP classification methods.
3. Detect repeated communication patterns that may indicate automated activity, periodic communication, or beaconing behaviour.
4. Identify unusual external IP addresses or unexpected communication

relationships.

5. Analyze commonly used and unusual ports and protocols.
6. Identify large data transfers or repeated data exchange patterns.
7. Highlight traffic behaviour that requires further security investigation.

For every finding:

- Explain why the observation is relevant from a security perspective.
- Provide evidence from the dataset.
- Separate confirmed observations from assumptions.
- Do not label traffic as malicious without sufficient evidence.
- Suggest validation steps a SOC analyst should perform.

At the end provide:

1. Overall security assessment.
2. Confirmed observations.
3. Potential concerns with confidence level.
4. Recommended next investigation steps.

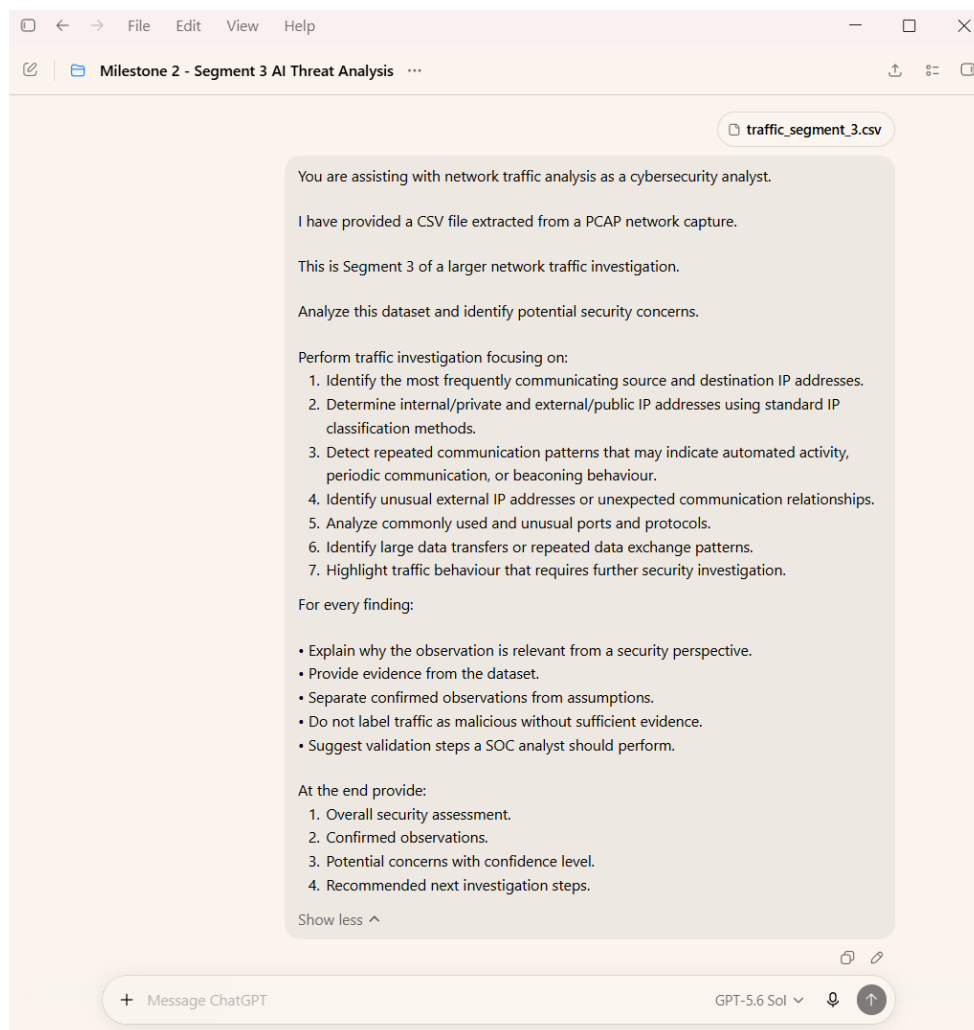


Figure 1: AI Prompt Submitted for Segment 3 Analysis

4. AI Generated Initial Findings (Segment 3)

Segment 3 Network Traffic Investigation Report

Dataset analysed: traffic_segment_3.csv

Investigation segment: Segment 3 of a larger PCAP network traffic investigation

The dataset was reviewed after correcting formatting issues found in the previous CSV version; the corrected file has properly formatted IP addresses and DNS-related rows.

Dataset Summary

Metric	Result
Total packets analysed	1,188
Unique source IP addresses	21
Unique destination IP addresses	23
Main protocols observed	TCP, TLS, HTTP, DNS, NBNS, ARP

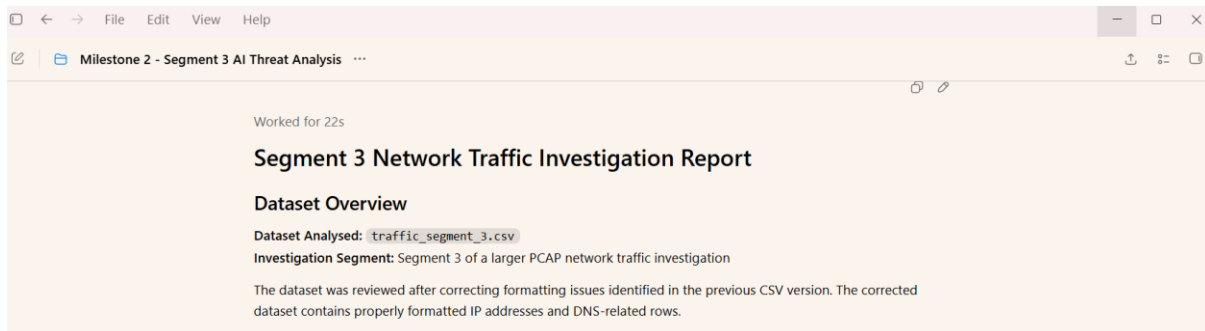


Figure 2: AI Generated Segment 3 Analysis Response

4.1 Most Frequently Communicating Source and Destination IP Addresses

Top Source IP Addresses

Rank	Source IP	Packet Count	Classification
1	10.10.0.33	616	Private/Internal
2	204.79.197.200	153	Public/External
3	204.11.56.48	68	Public/External
4	34.102.136.180	58	Public/External
5	8.8.8.8	44	Public/External

Top Destination IP Addresses

Rank	Destination IP	Packet Count	Classification
1	10.10.0.33	538	Private/Internal
2	204.79.197.200	165	Public/External
3	34.102.136.180	61	Public/External
4	204.11.56.48	54	Public/External
5	8.8.8.8	48	Public/External

10.10.0.33 is again the primary communicating host — it appears as the source in 616 packets and the destination in 538, and it talks to a wide range of public internet addresses. A single workstation reaching out to many external systems is normal in most environments, covering web browsing, software updates, cloud services, DNS resolution, and routine application traffic. The same shape of activity can also show up with automated background processes, malware communication, or bulk data transfer, so the behaviour still needs context before it can be written off either way.

4.2 Internal and External IP Address Classification

Private IP ranges follow the standard RFC 1918 definitions: 10.0.0.0/8, 172.16.0.0/12, and 192.168.0.0/16.

Internal / Private Address

IP Address	Classification	Reason
10.10.0.33	Private/Internal	RFC1918 private IPv4 range

Public / External Addresses Observed

IP Address	Classification
204.79.197.200	Public
204.11.56.48	Public
34.102.136.180	Public
8.8.8.8	Public
50.87.249.29	Public
35.214.181.99	Public
172.67.187.204	Public

10.10.0.33 communicates with all of these external hosts, which is expected for a normal client system. Outbound connections like this are worth a closer look mainly when the destination is unfamiliar, the connection frequency looks off, the connections repeat in an automated way, or the

data exchange pattern is abnormal — external communication on its own isn't evidence of anything malicious.

4.3 Repeated Communication Patterns and Beaconing Analysis

High-Frequency Communication Relationships

Communication	Outbound (10.10.0.33 →)	Inbound (→ 10.10.0.33)	Total
10.10.0.33 ↔ 204.79.197.200	165	153	318
10.10.0.33 ↔ 204.11.56.48	54	68	122
10.10.0.33 ↔ 8.8.8.8 (DNS, UDP/53)	48	44	92

Beaconing assessment

Repeated communication does exist between the internal host and several external services, but it doesn't show the traits of typical beaconing — fixed-interval callbacks, regular periodic timing, consistent C2-style traffic, or small repeated outbound messages. What's here looks more consistent with normal application communication, browser activity, cloud services, or a software update mechanism. Beaconing likelihood: low.

SOC validation steps

1. Identify the owner and device associated with 10.10.0.33.
2. Review endpoint activity — running processes, installed applications, browser activity, scheduled tasks, and startup entries.
3. Compare network activity with firewall logs, EDR alerts, and historical traffic patterns.

4.4 External IP Communication Requiring Review

External IP	Observation	Security Review Reason
204.79.197.200	Highest external communication volume	Confirm expected application/service
204.11.56.48	Frequent communication	Validate destination purpose
34.102.136.180	Repeated cloud-host communication	Confirm associated service
50.87.249.29	External web communication	Review reputation
35.214.181.99	Repeated external communication	Validate legitimacy
172.67.187.204	CDN/cloud-style communication	Confirm expected usage

The internal host reaches several internet destinations that likely belong to websites, cloud platforms, CDN infrastructure, application servers, or software update services. Nothing in the

dataset provides enough evidence to confirm malware infection, command-and-control communication, data theft, or unauthorized access.

4.5 Port and Protocol Analysis

TCP Traffic

Port	Packet Count	Common Usage
80	344	HTTP
443	222	HTTPS/TLS

UDP Traffic

Port	Packet Count	Purpose
53	48	DNS
137	2	NetBIOS Name Service

HTTP traffic on port 80 is common for ordinary web communication; since it isn't encrypted, any sensitive data transmission would be worth reviewing if it applies here. Port 443 traffic is expected for websites, secure applications, and cloud services. DNS communication on port 53 is normal for domain resolution and application connectivity — DNS can be abused for tunneling, malware communication, or data exfiltration, but nothing in this dataset points to that.

4.6 Large Frame Size and Data Exchange Analysis

Largest Frame Sizes Observed

Source IP	Destination IP	Frame Length
10.10.0.33	8.8.8.8	58,337 bytes
10.10.0.33	8.8.8.8	54,032 bytes
10.10.0.33	8.8.8.8	53,529 bytes
10.10.0.33	8.8.8.8	52,160 bytes

The largest frame values in the capture — up to 58,337 bytes — involve 10.10.0.33 talking to 8.8.8.8. That's worth checking against the original PCAP directly, since the extracted CSV fields may represent aggregated or reconstructed traffic rather than individual packet sizes; nothing here confirms data exfiltration. Unusual traffic volumes like this can point to ordinary application behaviour or file transfer activity, but frame size alone doesn't confirm DNS tunneling, exfiltration, or anything malicious.

SOC validation steps

4. Inspect the original PCAP for DNS query names and record types (A, AAAA, TXT, CNAME).
5. Look for very long or randomized domain names, high-frequency DNS queries, and suspicious payload patterns.

4.7 Traffic Behaviour Requiring Further Investigation

Finding	Evidence	Confidence
Main internal host communicating externally	10.10.0.33 communicating with multiple public IPs	Medium
High communication volume with 204.79.197.200	318 packets exchanged	Low-Medium
Large frame sizes involving DNS traffic	Up to 58 KB frame size	Low-Medium
Multiple external web connections	Several public destinations contacted	Low
Possible automated application activity	Repeated communication patterns	Low-Medium

Overall Security Assessment

Segment 3 traffic mainly looks like normal client-to-internet communication. 10.10.0.33 remains the system to watch, since it's talking to multiple external services over TCP, HTTP, HTTPS/TLS, and DNS. Nothing here directly confirms malware infection, command-and-control activity, data exfiltration, or unauthorized access — the behaviour should still be validated through endpoint investigation and by comparing it against the other PCAP segments.

Confirmed Observations

- Internal host identified: 10.10.0.33
- Internal-to-external communication observed
- Common protocols identified: TCP, HTTP, HTTPS/TLS, DNS, ARP
- Repeated communication relationships observed
- Large frame sizes detected

Potential Concerns and Confidence Levels

Concern	Confidence
Normal web/cloud communication	High
Automated application activity	Medium
Suspicious beaconing	Low

DNS tunnelling possibility	Low (requires packet payload validation)
Data exfiltration	Low

Recommended Next Investigation Steps

6. External IP reputation checks — run 204.79.197.200, 204.11.56.48, 34.102.136.180, and 50.87.249.29 through VirusTotal, AbuseIPDB, and other threat intelligence platforms.
7. DNS investigation — review query names, frequency, length, and record types.
8. Endpoint investigation — on 10.10.0.33, check running processes, installed software, browser activity, and persistence mechanisms.
9. Cross-segment correlation — since this is Segment 3 of a larger investigation, compare against the other segments for the same internal host behaviour, repeated external destinations, similar timing patterns, and consistent protocol usage.

Final Analyst Verdict

Overall risk rating: Low to Moderate

Segment 3 mainly shows expected internal-to-external network communication. 10.10.0.33 should stay the primary investigation target given its communication volume and repeated external interactions, but at this stage the evidence doesn't support classifying the traffic as malicious. Further endpoint analysis, DNS inspection, threat intelligence checks, and correlation with the other PCAP segments are recommended before making a final call.

5. Analyst Review of AI Output

The AI-generated findings were checked against the dataset directly.

Correct AI Observations

IP classification

It correctly identified 10.10.0.33 as private using the RFC1918 ranges — 10.0.0.0/8, 172.16.0.0/12, and 192.168.0.0/16.

Protocol identification

It also correctly identified TCP, HTTP, HTTPS/TLS, DNS, ARP, and NBNS in the traffic.

6. Limitations of AI Analysis

The AI's analysis was based on packet metadata only, without payload inspection, endpoint information, threat intelligence verification, or process-level visibility. Its findings should be treated as indicators, not confirmed threats.

7. Segment 3 Summary

Category	Observation
Primary internal host	10.10.0.33
Main external communication	204.79.197.200
Main protocols	TCP, HTTP, HTTPS, DNS
Beaconing	Not confirmed
Main concern	Repeated external communication
Risk level	Low-Medium, investigation required
Confirmed malware	No

8. Conclusion

ChatGPT picked out the important traffic patterns in Segment 3 — the internal host 10.10.0.33 and its several external communication relationships. Repeated communication and large frame values did show up, but the available metadata isn't enough to confirm malicious activity on its own. Packet payload analysis, DNS inspection, threat intelligence, and endpoint analysis are still recommended.